



CS-GY 6083 - B, Spring 2026 **Principles of Database Systems**

Project Part 1: 100 points with 20% weight

In this first part of the project, you will focus on designing a suitable relational database that can be used to store the data in the OLTP database system. In the second part of the project, you will then build a web-accessible frontend that will allow users to use the service via their browsers. Note that the second part of the project builds on top of this first part so you cannot skip this project part 1. You will not be allowed to change your team partner or change team project to individual or vice versa.

Project Guidelines:

In this first part of the project, you will design the relational database (using Oracle Data Modeler) that stores all the data about business case detailed below. You should use any relational database system (preferably MySQL) on your laptop or an internet accessible relational database on Azure/AWS cloud platform. Following is the business case and a list of steps for this part of the project. Note that in this first part, you will only deal with the database side of this project - a suitable web interface will be designed in the second part of the project.

Business Case:

National Insurance Care Excellence (NICE) is one of the largest life insurance provider companies in the USA. Recently NICE has expanded its operation to offer Auto and Home insurance to their customers. After careful analysis, the business team has decided to build new information systems for its Auto and Home insurance services. The project team has articulated a product scope to create a relational database that will incorporate schema design with all related entities, attributes, and appropriate relationships among entities to meet NICE's new business services.

The business analyst has provided following details/requirements for the database design.

- a) NICE intends to have all customers' details that include customer's full name, address, gender, marital status, and customer type. The possible data value of gender must be either "M", or "F" representing "Male" or "Female" respectively. A customer may choose not to provide gender data. The possible data value of marital status must be either "M", "S", or "W", representing "Married", "Single", and "Widow/Widower" respectively. The customer can be Automobile Insurance customer or Home Insurance customer or both. The customer type data value "A" represents Automobile Insurance customer and "H" represents home insurance customer.
- b) For Home Insurance customers, NICE intends to store home insurance policy start date and end date, home insurance premium amount, and home policy insurance status. If home insurance policy term is current, status column should have value "C", and if it is expired, it should have value "E". NICE wants to keep all previously expired policy data in the the same table.
- c) More than one home can be insured under each home insurance policy. For each home insured, NICE intends to store home purchase date, home purchase value, home area in Sq. Ft., Type of home (one of the value as, S,M,C,T representing Single family, Multi Family, Condominium, Town house respectively). To decide appropriate home insurance premium, NICE intending to store four parameters namely: Auto Fire notification, Home Security System, Swimming Pool, and Basement. Please note following possible values for each of the parameters.

Parameter	Possible Value	Meaning
Auto Fire Notification	1	There is automatic fire notification to the fire department
Auto Fire Notification	0	There is NO fire notification to the fire department
Home Security System	1	The home security system is installed and monitored
Home Security System	0	The home security system is not installed or not monitored
Swimming Pool	"U"	Underground swimming pool
Swimming Pool	"O"	Overground swimming pool

Swimming Pool	"I"	Indoor swimming pool
Swimming Pool	"M"	Multiple swimming pool
Swimming Pool	null	No swimming pool
Basement	1	There is a basement
Basement	0	There is NO basement

- d) Each home insurance policy will generate one or more invoices on different invoice date. Invoice should include payment due date, and invoice amount.
- e) NICE accepts home insurance premium in installments and each payment should have stored with payment date and method of payment. NICE accepts payment either as "PayPal", "Credit", "Debit", "Check"
- f) For Auto Insurance customers, NICE intends to store auto insurance policy start date and end date, auto insurance premium amount, and auto policy insurance status. If auto insurance policy term is current, status column should have value "C", and if it is expired, it should have value "E". NICE wants to keep all previously expired policy data in the same table.
- g) Each Auto insurance policy will generate one or more invoices on different invoice dates. Invoice should include payment due date, and invoice amount.
- h) NICE accepts Auto insurance premium in installments and each payment should be stored with payment date and method of payment. NICE accepts payment either as "PayPal", "Credit", "Debit", "Check"
- i) For each of Auto Insurance policy, one or more vehicles can be insured. For each insured vehicle, NICE intends to store vehicle VIN (vehicle identification number), vehicle make-model-year, and status of the vehicle. The status of the vehicle can be one of "L", "F", or "O" representing "Leased", "Financed", and "Owned".
- j) For each vehicle insured, NICE intends to store details of drivers. There may be more than one driver for each vehicle insured. For each driver, NICE intends to store driver's license number, name, and age.

For this project, identify appropriate entities, identify relationships among entities, attributes of each entity and their datatype- size-constraints-comments. Applying database design principles develop and implement a centralized relational database system that collects relevant data for NICE.

For any business rule not defined, please state all valid assumptions you have made (if any) while designing and implementing the database.

All table names must have prefix made of first character of first name of each team member, e.g., if team members first names are Peter, Charlie, and Geroge then table names should have prefix PCG. In that case table names should be PCG_CTUSER, PCG_PAYMENT etc.

Submission:

Submit following deliverables towards project part 1 assignment

- a) Screenshots of Logical model and Relational model. Please state all the valid assumptions you have made, if any, in support of your data model. If you have not made any assumptions, state that NO ADDITIONAL ASSUMPTIONS ARE MADE [30 points]
- b) Apply at least five different types of CHECK constraints to apply defined business rules to enforce data consistency. Submit DDL code of check constraints. [20 points]
- c) Populate meaningful sample data for all entities (at least 10 to 30 records per entity). List total number of records populated for each entity (just record counts, not full data set). Submit COUNT (*) query and result for each table. Any table with less than 10 records will be subject to point deductions. [20 points]
- d) Write data dictionary queries that detail all tables, columns-data type-size-mandatory/optional, constraints, and attribute comments of schema objects. Submit data dictionary queries and screenshots of their corresponding results. Please note that, if you have large records in results, you will need to take multiple screenshots. Meaning that, you will need to submit screenshots of the full set of results and not the partial. Partial screenshot will be subject to point deductions. [20 points]
- e) DDL code of all Tables [10 points]

Submission should be a single PDF document with naming convention <Prefix_Project_Part1.pdf>, e.g. PCG_Project_Part1.pdf, if your group's initial is PCG.

- On the cover page clearly state course, section, semester term, submission date, name and Net-id all team members.
- There should be a page for the table of contents after the cover page.
- Write work distribution among team members (who has worked on which part of the project).
- Write team reflection on project work (what have you learned, challenges faced, and how challenges are addressed and solved).
- Write individual reflection about what you have learned from the project. All team members to write individual reflection in the same single document.

Each student of the project team needs to submit the documentation on Brightspace. Any submission by email will be rejected, unless there is a legit reason and approved by the professor.

Grading Rubrics

- -2 points for each type of design principal violation against DB DESIGN CHECK LIST posted on course website
- -2 points for invalid table/attribute names (multiple/reserve words, tables without team prefix, over 30 characters)
- -2 ERD are not fully expanded
- -5 mismatch between Logical and Relational ERD
- -5 missing Logical or Relational model
- -5 if any assumptions are made (other than stated business rules), and they are not stated
- -4 for each of missing check constraints
- -10 if enough number of records are not populated for any table
- -10 if data dictionary information (queries and their results) is incomplete
- -20 if data dictionary information (queries and their results) is not submitted
- -10 if DDL code is not submitted, or partially submitted
- 10% penalty if incorrect document is submitted and corrected within grace period
- 10% late submission penalty, if assignment submitted before the grace period
- 10% late submission penalty, if work distribution is not provided
- 10% late submission penalty, if team reflection is not provided
- 10% late submission penalty, if individual reflection is not provided (this applied to only student(s) of the team who have not provided individual reflection)
- -100 if involved in plagiarism
- -100 if assignment submitted after grace period (24 hours of assignment deadline)

===== END OF THE PROJECT PART 1 DOCUMENT =====